Equations (3.60) and (3.61) are called as **cyclotron frequency or gyro-frequency**.

From equations (3.59), (3.60) and (3.61), we infer that time period and frequency depend only on charge-to-mass ratio (specific charge) but not on velocity or the radius of the circular path.

If a charged particle moves in a region of uniform magnetic field such that its velocity is not perpendicular to the magnetic field, then the velocity of the particle is split up into two components; one component is parallel to the field while the other component perpendicular to the field. The component of velocity parallel to field remains unchanged and the component perpendicular to the field keeps changing due to Lorentz force. Hence the path of the particle is not a circle; it is a helical around the field lines as shown in Figure 3.46.

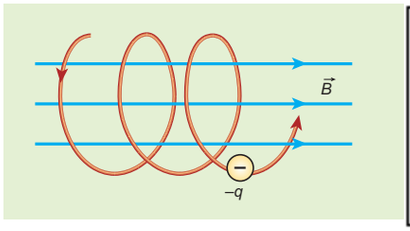


Figure 3.46 Helical path of the electron in a uniform magnetic field

For an example, the helical path of an electron when it moves in a magnetic field is shown in Figure 3.47. Inside the particle detector called cloud chamber, the path is made visible by the condensation of water droplets.

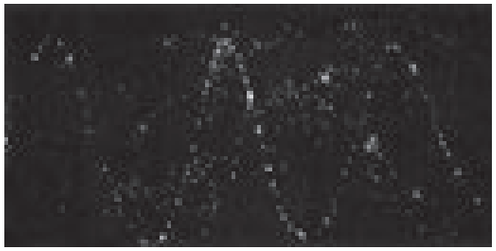


Figure 3.47 Helical path of the electron inside the cloud chamber

EXAMPLE 3.19

An electron moving perpendicular to a uniform magnetic field 0.500 T undergoes circular motion of radius 2.50 mm. What is the speed of electron?

Solution

Charge of an electron $q = -1.60 \times 10^{-19}$ C

$\Rightarrow |q| = 1.60 \times 10^{-19}$ C

Magnitude of magnetic field $B = 0.500$ T

Mass of the electron, $m = 9.11 \times 10^{-31}$ kg

Radius of the orbit, $r = 2.50$ mm $= 2.50 \times 10^{-3}$ m

Speed of the electron, $v = |q| \dfrac{rB}{m}$

$$v = 1.60 \times 10^{-19} \times \frac{2.50 \times 10^{-3} \times 0.500}{9.11 \times 10^{-31}}$$

$$v = 2.195 \times 10^{8} \text{ ms}^{-1}$$

EXAMPLE 3.20

A proton moves in a uniform magnetic field of strength 0.500 T magnetic field is directed along the x-axis. At initial time, $t = 0\ s$, the proton has velocity $\vec{v} = \left(1.95 \times 10^{5}\,\hat{i} + 2.00 \times 10^{5}\hat{k}\right)\text{ms}^{-1}$. Find

(a) At initial time, what is the acceleration of the proton.

(b) Is the path circular or helical? If helical, calculate the radius of helical trajectory and also calculate the pitch of the helix (Note: Pitch of the helix is the distance travelled along the helix axis per revolution).

Solution

Magnetic field $\vec{B} = 0.500\hat{i}$ T

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